EXPLAINABLE REVIEWED-ITEM RETRIEVAL: SEARCHING RESTAURANTS BY TASTE PREFERENCE

by

Chidsanuphong Pengchai (st127004)
Samichi Rungta (st127012)

Group: Flower

Asian Institute of Technology
Department of Data Science and Artificial Intelligence
Thailand

September 17, 2026

CONTENTS

1 Introduction **5**
- 1.1 Background of the Work . . . 5
- 1.2 Why Conduct This Project . . . 6
- 1.3 Business and Research Understanding of the Problem . . . 6
- 1.4 Potential Impact and Benefits . . . 7

2 Problem Statement **8**
- 2.1 Problem Definition . . . 8
- 2.2 Objectives . . . 8
- 2.3 Scope and Boundaries . . . 9

3 Related Works **11**
- 3.1 Reviewed-Item Retrieval and the Fusion Gap . . . 11
- 3.2 From Collaborative Filtering to Cold-Start, Content-Based Recommendation . . . 11
- 3.3 Natural-Language Preference Elicitation at the Review-Sentence Level 12
- 3.4 Lexical and Dense Text Retrieval . . . 12
- 3.5 Explainable Recommendation . . . 12
- 3.6 Aspect-Based Sentiment and the Flavour-Feature Layer . . . 13
- 3.7 Thai Language Processing . . . 13

4 Datasets **14**
- 4.1 RIRD: Reviewed-Item Retrieval Dataset (Primary, Labelled) . . . 14
- 4.2 Wongnai Reviews (Thai, Unlabelled: Demonstration Only) . . . 14
- 4.3 Thailand Foods: Dish Gazetteer (Supporting, Optional Ablation) . . . 15
- 4.4 M-ABSA Thai: Aspect-Based Sentiment (Supporting, Optional Ablation) . . . 15
- 4.5 Justification and Complementarity . . . 15

5 Methodology **16**
- 5.1 Exploratory Data Analysis . . . 16
- 5.2 Data Preprocessing . . . 17
- 5.3 Proposed Model(s) . . . 17

6 Baseline and Proposed Approach **19**
- 6.1 Baseline Models . . . 19
- 6.2 Proposed Approach . . . 19
- 6.3 Expected Improvement Over Existing Solutions . . . 20
- 6.4 Unique Features and User Benefits . . . 20

7 Preliminary Results **22**
7.1 RIRD: Feasibility as the Evaluation Backbone 22
7.2 PMD Relevance Labels: Confirming the Evaluation Protocol is Usable 23
7.3 Wongnai: Feasibility and Limits of the Thai Demonstration 24
7.4 Thailand Foods and M-ABSA: Feasibility as Supporting Resources . . 26
7.5 Baseline Model Results 27
7.5.1 BM25 27
7.5.2 Dense Retrieval Without Fusion 28
7.5.3 Cross-Model Comparison 29
7.6 Summary 30

LIST OF FIGURES

7.1 Distribution of review star ratings in RIRD. 22
7.2 Distribution of RIRD review length in words (clipped at the 99th percentile). 23
7.3 Number of annotators (out of five) voting a query–restaurant pair relevant. 24
7.4 Distribution of Wongnai star ratings (0–4 scale). 25
7.5 Distribution of Wongnai review length in words (clipped at the 99th percentile). 25
7.6 Thailand Foods dish counts by meal course. 26
7.7 Sentiment polarity distribution in M-ABSA Thai (food and restaurant domains). 27
7.8 Cross-model comparison of BM25 and dense (no-fusion) baselines on RIRD across MAP, nDCG@5/10, and Recall@5/10. 29

LIST OF TABLES

6.1 Comparison of existing solutions against the proposed approach. . . . 20

7.1 BM25 baseline results on RIRD (100 queries, 50 restaurants). 28
7.2 Dense (no-fusion) baseline results on RIRD (100 queries, 50 restaurants). 28
7.3 Cross-model comparison on RIRD. 29

CHAPTER 1

INTRODUCTION

1.1 Background of the Work

Food occupies a central place in Thai culture and in Thailand's national economy, and Thai cuisine—street food in particular—is treated as an instrument of national soft power that drives tourism revenue. The global standing of this food scene is measurable: Bangkok was ranked the world's No. 2 food city (and No. 1 in Asia) in *Time Out*'s *Best Cities for Food 2026*, behind only Lima, with an 80% expert-panel score and 81% of surveyed locals rating the city highly for food quality and diversity. The street-food segment is not a marginal niche but the backbone of the sector: Thailand has roughly 103,000 street-food operators, about 69% of all food businesses, generating approximately 271,355 million baht—about 32.4% of food-business revenue—making street food the single largest foodservice segment. Food is also where visitors spend: food and beverage accounted for 23% of foreign tourists' spending in 2023 according to the Bank of Thailand's Q2 2024 Monetary Policy Report (second only to accommodation at 36%). What distinguishes Thai food choice from many other cuisines is the centrality of *flavour balance*—the interplay of spicy, sour, sweet, salty, and rich (fatty/umami) notes—so that diners frequently choose a meal by desired taste profile rather than by dish name.

The digital layer over this landscape is dominated by two incumbents. Wongnai (now part of the LINE MAN Wongnai group) is Thailand's leading restaurant-review platform; its app draws on user-generated reviews and feedback from over 5.7 million users, with more than 2.4 million reviews and 84 million photos across more than one million locations nationwide. Google Maps provides the other major review and discovery surface. Both, however, expose discovery primarily through keyword search, category filters, star ratings, and geographic proximity—none of which lets a user search by the *sensory experience* they want. Academic NLP has matured enough to make such a system feasible: transformer sentence encoders (Devlin et al., 2019; Reimers & Gurevych, 2019) and multilingual models, together with Thai-specific pretrained models such as WangchanBERTa (Lowphansirikul et al., 2021) and the publicly released Wongnai review corpus, provide the ingredients for processing

review text—including low-resource Thai—at scale.

1.2 Why Conduct This Project

Three gaps motivate the work. First, **there is no mainstream way to search restaurants by taste preference**: a user who wants "something sour and spicy that is not expensive" must translate that intent into keywords, guess at dish names, and manually cross-read reviews. Second, the recommendation **cold-start problem** is acute here—a tourist or newcomer has no interaction history and no ratings, so collaborative-filtering systems that depend on behavioural logs (Koren et al., 2009) cannot serve them, and the natural-language, content-based paradigm is the appropriate alternative (Gao et al., 2021; Kostric et al., 2024). Third, the evidence needed to answer such a query **already exists latent in review text**—diners constantly describe spiciness, sourness, value, and atmosphere in their own words—yet this signal remains untapped as a first-class retrieval mechanism, and prior work leaves open *how* that sentence-level evidence should be combined into a restaurant-level decision (Abdollah Pour et al., 2023; Zhang & Chen, 2020).

1.3 Business and Research Understanding of the Problem

The problem is felt by several groups. **Tourists and newcomers** bear a search cost: they cannot express taste intent as keywords and cannot efficiently read long review threads, leading to poor or risky choices. **Locals** face choice overload across a very large restaurant catalogue. **Vendors**—especially small and street-food operators—are poorly served by rating-and-proximity ranking, because a distinctive but niche flavour offering is invisible to a user who cannot search for it. Crucially, star ratings measure *overall satisfaction*, not spicy/sour/sweet/rich intensity, so the incumbents' interaction model captures none of this. The research question is therefore not "which restaurant is best" but a **retrieval question**: given a free-text preference query, can review text be used to retrieve and rank restaurants in a way that is both accurate and self-justifying, and does the way sentence-level evidence is *fused* into a restaurant-level score materially affect that accuracy? This reframing turns a subjective "flavour recommender" into a measurable information-retrieval problem with human relevance judgements (Abdollah Pour et al., 2023).

1.4 Potential Impact and Benefits

For **tourists and newcomers**, the system removes the language and knowledge barrier by accepting intent in natural language and by returning, alongside each recommendation, the exact review sentences that justify it—an explainable output that builds trust (Zhang & Chen, 2020). For **locals** it offers taste-driven discovery beyond rating and distance. For **vendors** it creates a fairer surface where a distinctive flavour profile can be matched to demand rather than buried beneath aggregate stars. Methodologically, and in line with the scope fixed in Chapter 2, the project contributes: (i) a *systematic comparison of sentence-to-restaurant fusion functions* (max, mean-of-top-k, rating-weighted sum), evaluated on the labelled RIRD benchmark with nDCG@10, MAP, and Recall@10—evidence currently missing because the anchor prior work reports only a single aggregation method (top-K mean) and only two metrics (R-Precision, MAP) (Abdollah Pour et al., 2023); (ii) an explainable, content-based retrieval pipeline that attaches justifying review sentences to every ranked restaurant (Zhang & Chen, 2020; Kostric et al., 2024); (iii) a *cross-lingual demonstration* of the same language-agnostic pipeline on the Thai Wongnai corpus, showing the method transfers to a low-resource language at the review/sentence level; and (iv) an optional *flavour-feature ablation* that tests whether explicit sensory features add value over pure dense retrieval, judged against independent human labels rather than against its own domain assumptions.

CHAPTER 2

PROBLEM STATEMENT

2.1 Problem Definition

We address a natural-language, review-grounded restaurant retrieval problem. Let $R = \{r_1, \ldots, r_n\}$ be a corpus of restaurants, where each restaurant r_i is associated with a set of reviews $D_i = \{d_{i,1}, \ldots, d_{i,m}\}$, and each review is decomposed into sentences $S_i = \{s_{i,1}, s_{i,2}, \ldots\}$. Given a free-text preference query q, the task is to produce a **ranking of the restaurants in R by relevance to q**. The system first scores each sentence against the query with a sentence encoder, $\text{score}(q, s_{i,j})$; it then applies a fusion function f that aggregates the sentence-level scores of restaurant r_i into a single restaurant-level score

$$\text{Score}(q, r_i) = f\big(\{\, \text{score}(q, s_{i,j}) : s_{i,j} \in S_i \,\}\big), \tag{2.1}$$

used for ranking; the top-ranked restaurants are returned together with their highest-scoring justifying sentences. The central research contribution concerns the choice of f—whether *max*, *mean-of-top-k*, or *rating-weighted sum* yields the best ranking. This is the *reviewed-item retrieval* formulation, in which the two-level restaurant/review structure requires an explicit aggregation step (Abdollah Pour et al., 2023).

The formulation is language-agnostic. It is *developed and quantitatively evaluated* on the English RIRD benchmark (Abdollah Pour et al., 2023), which supplies the restaurant grouping, 100 preference queries, and human relevance labels needed to measure ranking quality. The same pipeline is then *demonstrated* on the Thai Wongnai corpus—for example, on the query "I want something sour and spicy that is not expensive"—at the review/sentence level, illustrating cross-lingual transfer to a low-resource setting where relevance labels are not yet available.

2.2 Objectives

1. **Implement the retrieval pipeline.** Build sentence-level dense retrieval using transformer sentence embeddings (Reimers & Gurevych, 2019; Karpukhin et al.,

2020): split reviews into sentences, encode query and sentences into a shared vector space, and retrieve the top-k sentences per query.

2. **Implement and systematically compare fusion.** Compare fusion strategies (max, mean-of-top-k, rating-weighted sum) for aggregating sentence scores into a restaurant score—the primary research contribution, filling the gap left by the anchor work, which tested only the top-K mean (Abdollah Pour et al., 2023).

3. **Benchmark against baselines on RIRD.** Compare the pipeline against random ranking, rating-only sorting, BM25 lexical matching (Robertson & Zaragoza, 2009), and embedding retrieval with mean fusion, reporting **nDCG@10, MAP, and Recall@10** on the RIRD relevance labels.

4. **Deliver explainable output.** For every recommended restaurant, return the specific review sentences that justify its position in the ranking (Zhang & Chen, 2020; Kostric et al., 2024).

5. **Demonstrate cross-lingual application (Thai).** Port the same pipeline to the Thai Wongnai corpus and show qualitative review/sentence-level retrieval for Thai preference queries. This is a *demonstration*, not a measured claim: Wongnai lacks restaurant grouping, queries, and relevance labels, so no quantitative result is asserted on it.

6. **Test the flavour-feature layer (optional ablation).** Evaluate whether adding explicit sensory-intensity features (built from the Thailand Foods gazetteer and Thai aspect signals) improves ranking over pure dense retrieval, judging the layer against **independent human labels** rather than against its own domain assumptions (Pontiki et al., 2014).

2.3 Scope and Boundaries

The project is deliberately scoped to **content- and review-based retrieval**. All *quantitative* claims are made on the labelled RIRD benchmark; the Thai Wongnai corpus is used only for a *qualitative* cross-lingual demonstration, and full Thai *restaurant-level* recommendation—which would require Wongnai metadata with restaurant identifiers or a purpose-built labelled Thai query set—is left as future work. The project does **not** attempt collaborative filtering; it does **not** use click logs or any behavioural interaction data; and it does **not** build per-user profiles. It is **single-turn** (one query

in, one ranking out) rather than a multi-turn conversational dialogue. It does not perform menu-price extraction, delivery logistics, or real-time availability checking, and it does not *generate* new natural-language explanations—only *select* existing review sentences as evidence. Geographic and temporal personalisation are out of scope. These boundaries keep the study focused on the measurable retrieval-and-fusion research question defined above.

CHAPTER 3

RELATED WORKS

3.1 Reviewed-Item Retrieval and the Fusion Gap

The closest prior work and the anchor for this project is the Reviewed-Item Retrieval (RIR) task of Abdollah Pour et al. (2023), which formalises exactly our setting: given a natural-language query and items each described by many reviews, a system must aggregate query–review similarities into an item-level score and rank items. The authors distinguish *early fusion* (average an item's reviews into one embedding, then score) from *late fusion* (score each review, then aggregate the top-K), and release the RIRD dataset—50 Toronto restaurants, roughly 29,000 reviews, and 100 preference queries in five categories (indirect, negation, general, detailed, contradictory) with binary relevance from five annotators. Their best late-fusion contrastive model reaches MAP 0.626, well above BM25 and off-the-shelf BERT. Crucially for us, the paper evaluates with *only* R-Precision and MAP and aggregates with *only* a top-K mean; it does not test max or rating-weighted fusion, nor report nDCG or Recall. This is the precise, non-manufactured gap our fusion comparison fills, on the same dataset and query protocol.

3.2 From Collaborative Filtering to Cold-Start, Content-Based Recommendation

The dominant classical recommendation paradigm is collaborative filtering via matrix factorization (Koren et al., 2009), which learns latent user and item factors from a history of ratings or interactions. This paradigm is powerful when behavioural logs are abundant, but it is structurally unable to serve the cold-start user—a tourist or newcomer with no history—which is precisely our target scenario. Surveys of conversational and content-based recommendation (Gao et al., 2021) argue that natural-language interfaces let users express current, context-specific intent directly and are especially valuable when interaction data is unavailable. Our system is a single-turn instance of this natural-language paradigm: it deliberately avoids collaborative filtering and instead retrieves over review content, which is exactly what makes it

applicable to users the classical paradigm cannot reach.

3.3 Natural-Language Preference Elicitation at the Review-Sentence Level

Kostric et al. (2024) observe that ordinary users often cannot answer questions about item *attributes*—they know how they intend to use an item, not its technical specifications—and show that review sentences are a mineable source of user-facing, preference-relevant language. This supports two of our design decisions. First, it validates operating at the *review-sentence* granularity, the same level at which we retrieve and return justifications. Second, its central premise matches our tourist/newcomer use case exactly: a visitor cannot name a Thai dish but can type "something sour and spicy, not expensive," which is why free-text taste queries are preferable to fixed attribute filters.

3.4 Lexical and Dense Text Retrieval

Our retrieval stack builds on two families of methods. The lexical family is anchored by BM25 (Robertson & Zaragoza, 2009), the probabilistic relevance model that remains a strong and standard baseline; we use it as the lexical baseline our embedding approach must beat. The dense family begins with BERT (Devlin et al., 2019) and its sentence-level adaptation, Sentence-BERT (Reimers & Gurevych, 2019), which produces semantically meaningful sentence embeddings comparable by cosine similarity—the technique we use to encode both queries and review sentences into a shared space. Dense Passage Retrieval (Karpukhin et al., 2020) then established that dense embedding retrieval can substantially outperform lexical matching for natural-language queries; our BM25-vs-dense comparison on restaurant reviews is a structurally identical experiment applied to a new domain.

3.5 Explainable Recommendation

The explainable component of our system—returning the specific review sentences that justify each restaurant's ranking—sits squarely within the "review-sentence-as-explanation" paradigm catalogued by Zhang & Chen (2020)'s survey of explainable recommendation. The survey organises the field by information source and model

type, and argues that user reviews are a rich explanation source because individual sentences can both justify a recommendation and surface the aspects a user cares about. It also stresses evaluating explanations against independent signals rather than self-referentially—guidance we adopt for the optional flavour-feature layer.

3.6 Aspect-Based Sentiment and the Flavour-Feature Layer

The optional flavour layer draws on aspect-based sentiment analysis (ABSA), defined as a mainstream task by Pontiki et al. (2014), whose restaurant-domain taxonomy—*food, service, price, ambience, anecdotes/miscellaneous*—the multilingual M-ABSA resource (used in our project) inherits. This citation lets us state a precise design boundary: the standard ABSA taxonomy contains *no* flavour-intensity axes such as spicy, sour, or sweet. Consequently, off-the-shelf ABSA aspects cannot serve as ground truth for a flavour-intensity layer; the layer must instead be validated against independent human labels. This both motivates the layer's place as an optional ablation and directly answers the concern that ingredient- and lexicon-derived flavour features might otherwise validate each other circularly.

3.7 Thai Language Processing

Because the cross-lingual demonstration operates on Thai text, which is written without inter-word spaces and is comparatively low-resource, it relies on Thai-specific language technology. WangchanBERTa (Lowphansirikul et al., 2021) is a transformer model pretrained on large Thai corpora and, together with multilingual sentence encoders and the PyThaiNLP toolkit for word segmentation, makes review-level retrieval over the Wongnai corpus feasible without English-centric assumptions.

CHAPTER 4

DATASETS

4.1 RIRD: Reviewed-Item Retrieval Dataset (Primary, Labelled)

The primary and sole source of quantitative evaluation in this project is RIRD, released alongside the anchor Reviewed-Item Retrieval paper (Abdollah Pour et al., 2023). It consists of two linked files. The review file, `50_restaurants_all_rates.csv`, contains 29,168 reviews across 7 columns (restaurant identifier, reviewer identifier, star rating, review text, restaurant name, cuisine category, and date), covering 50 Toronto restaurants with dates spanning 2008–2019. The second file, `PMD.csv`, provides 100 free-text natural-language preference queries evaluated against all 50 restaurants, yielding 5,000 query–restaurant pairs after one exact duplicate row is removed. Each pair carries binary relevance judgements from five independent human annotators together with an aggregated label. The aggregated binary relevance label is the target variable used throughout Chapter 6 for computing nDCG@10, MAP, and Recall@10.

4.2 Wongnai Reviews (Thai, Unlabelled: Demonstration Only)

To support the cross-lingual demonstration described in Objective 5, we use the Wongnai review corpus (Highgroundbkk, n.d.), comprising 40,000 training and 6,203 test Thai-language restaurant reviews, each with only two fields: review text and a 0–4 star rating. This dataset carries no restaurant identifiers and no query-relevance labels, so it cannot support quantitative ranking evaluation; it is used exclusively to demonstrate that the trained pipeline transfers to Thai text at the review/sentence level, consistent with the scope boundary fixed in Section 2.3.

4.3 Thailand Foods: Dish Gazetteer (Supporting, Optional Ablation)

The optional flavour-feature ablation (Objective 6) draws on a curated catalogue of Thai dishes (Sodchun, n.d.), giving English name, Thai script name, key ingredients, meal course, province, and region for each dish. This dataset carries no target variable; it functions as a lookup gazetteer for normalising food-term mentions when constructing the optional sensory-feature layer.

4.4 M-ABSA Thai: Aspect-Based Sentiment (Supporting, Optional Ablation)

We additionally draw on the Thai subset of the multilingual M-ABSA dataset (Multilingual-NLP, n.d.), restricted to its `restaurant` and `food` domains, which provides sentence-level Thai text annotated with coarse aspect categories and sentiment polarity, inheriting the SemEval-2014 restaurant taxonomy (Pontiki et al., 2014). As discussed in Section 3.6, this taxonomy contains no flavour-intensity axes, so it cannot serve as ground truth for a flavour layer; it is used only as a complementary sentiment signal when validating the optional ablation against sources independent of our own feature construction.

4.5 Justification and Complementarity

No single dataset satisfies every requirement of the project, but the four sources divide the labour cleanly. RIRD is the only dataset with independent human relevance judgements and is therefore the sole basis for every measured claim in Chapter 6, directly addressing the concern that manually-derived features might otherwise be validated circularly against themselves. Wongnai supplies the volume of real Thai text needed to demonstrate cross-lingual transfer without requiring translation. The dish gazetteer and M-ABSA Thai support the optional flavour ablation without being asked to carry any evaluation weight they cannot support.

CHAPTER 5

METHODOLOGY

5.1 Exploratory Data Analysis

EDA will be conducted separately on the English (RIRD) and Thai (Wongnai, Thailand Foods, M-ABSA) sides of the project, since the two halves serve different roles: RIRD is the measured evaluation set, while the Thai corpora are inspected primarily to confirm what kind of demonstration they can support.

RIRD reviews. We will examine the review-count distribution per restaurant, the star-rating distribution, and the review-length distribution, together with basic missing-value and duplicate-row checks. A language-composition check using a Unicode-range script detector will confirm whether review text and queries are English, Thai, or mixed, which determines whether RIRD can serve as a Thai evaluation source or only as the English backbone.

RIRD relevance labels (PMD). We will inspect the balance of positive versus negative relevance labels overall and across the five query categories (indirect, negation, general, detailed, contradictory), examine per-annotator positive rates, and compute inter-annotator agreement to identify any low-consensus queries worth flagging before evaluation.

Wongnai. We will examine the star-rating distribution (noting the 0–4 scale differs from RIRD's 1–5 scale), the review-length distribution, and the proportion of genuinely Thai versus code-switched text. We will also confirm what restaurant- or query-level metadata, if any, is present, since this determines whether Wongnai can support anything beyond a review/sentence-level demonstration.

Thailand Foods. We will check dish counts by region and course, screen for duplicate dish entries, and inspect ingredient-vocabulary coverage to assess its usefulness as a query-normalisation gazetteer.

M-ABSA Thai. We will examine the distribution of aspect categories and sentiment polarity within the food and restaurant domains, checking specifically whether any category aligns with flavour-intensity concepts (spicy, sour, sweet) rather than only general sentiment.

5.2 Data Preprocessing

Missing values. Missing `categories` values in RIRD will be filled with "Unknown"; any duplicate rows identified in `PMD.csv` during EDA will be dropped prior to evaluation. Placeholder values (*Various*, *Unknown*) in the gazetteer's region field will be retained as legitimate categories rather than treated as missing. M-ABSA parsing will drop any line that fails to yield a well-formed aspect-sentiment tuple.

Outlier analysis. Very short RIRD reviews will be flagged as possible spam or near-empty entries and retained unless entirely blank. Should the PMD positive rate prove low, as anticipated, this will motivate rank-aware metrics (nDCG, MAP, Recall@10) over accuracy-style metrics, which would be misleading under such imbalance. Wongnai's non-standard 0–4 rating scale will require no transformation, as it is used only as an auxiliary filter in the demo, never as a regression target.

Distribution and skewness. Review lengths in both RIRD and Wongnai are expected to be right-skewed. Rather than transform this away, we address it structurally: every review will be split into sentences prior to encoding, so the unit of retrieval is the sentence rather than the full review, limiting the influence of a small number of very long reviews on any single restaurant's score.

Feature preparation. RIRD review text will be split into sentences with the NLTK sentence tokenizer. Wongnai text will be segmented with the PyThaiNLP sentence segmenter, since Thai is written without inter-word spaces. All sentences and queries will be encoded into a shared vector space using pretrained sentence-transformer models (Reimers & Gurevych, 2019); no classical feature scaling will be applied, as these embeddings are already unit-normalised prior to cosine similarity computation. The deduplicated Thailand Foods gazetteer will be tokenised into keyword lists to normalise food-term mentions in incoming queries for the optional ablation.

5.3 Proposed Model(s)

The core model is sentence-level dense retrieval with an explicit fusion step (Equation 2.1): restaurant reviews are split into sentences, every sentence and query is encoded with a pretrained sentence encoder, sentence-level relevance is scored by cosine similarity, and per-sentence scores are aggregated into a restaurant-level score by a fusion function f. In line with the "basics over complexity" guidance received on this proposal, the comparison is deliberately run along two independent axes rather

than one:

- **Fusion function** — max pooling, mean-of-top-k ($k \in \{3, 5, 10\}$), and a rating-weighted sum, as introduced in Chapter 6.
- **Sentence encoder** — a compact English-only encoder (`all-MiniLM-L6-v2`), the multilingual encoder already required for the Thai demonstration (`paraphrase-multiling` and a retrieval-tuned multilingual alternative (`multilingual-e5-base`).

Crossing these two axes produces a small grid of encoder–fusion combinations, each evaluated identically against the RIRD relevance labels. This adds a second, low-complexity axis of comparison without requiring any model training: every encoder is used frozen and off-the-shelf, and the experiment is a direct application of embedding and similarity-scoring fundamentals rather than a more elaborate architecture.

CHAPTER 6

BASELINE AND PROPOSED APPROACH

6.1 Baseline Models

Four baselines anchor the comparison. **Random ranking** and **rating-only sorting** establish the lower bound achievable without using review content at all. **BM25** (Robertson & Zaragoza, 2009) is the standard lexical baseline, scoring review sentences by term-frequency–inverse-document-frequency weighting and aggregating to a restaurant-level score; it is the same baseline used in the anchor RIR paper (Abdollah Pour et al., 2023), making our numbers directly comparable. **Dense retrieval without fusion** mean-pools all of a restaurant's sentence embeddings into a single vector before ranking by cosine similarity to the query, isolating what the fusion step in the proposed model contributes beyond simply using embeddings at all.

6.2 Proposed Approach

The proposed system performs sentence-level dense retrieval with an explicit fusion step, as formalised in Equation 2.1. Reviews are split into sentences; sentences and query are encoded into a shared space (Devlin et al., 2019; Reimers & Gurevych, 2019); the top-k most relevant sentences per query are retrieved by cosine similarity; per-sentence scores are fused into a restaurant-level score; and restaurants are ranked by that fused score, with their top-scoring sentences attached as justification. This comparison is run over three fusion functions crossed with three sentence encoders, and the combination achieving the best nDCG@10/MAP on the RIRD evaluation set is reported as the final proposed method, together with the full grid to make the individual contribution of encoder choice and fusion choice separately visible.

6.3 Expected Improvement Over Existing Solutions

Table 6.1: Comparison of existing solutions against the proposed approach.

Dimension	**Existing solutions (Kaggle / Hugging Face)**	**Proposed approach**
Query type	Collaborative filtering or tag-based search; no support for free-text preference queries	Explicitly evaluated on indirect, negated, and contradictory queries (Abdollah Pour et al., 2023)
Explainability	Ranked lists with no justification	Every ranking is accompanied by the specific review sentences that support it (Zhang & Chen, 2020)
Language	Thai restaurant search tools are keyword-only	Same multilingual pipeline handles both English evaluation and Thai demonstration without a translation step
Evaluation	Star-rating prediction accuracy is the typical reported metric	Ranking quality is evaluated with nDCG@10, MAP, and Recall@10 against human-annotated relevance (Abdollah Pour et al., 2023)
Fusion	The anchor prior work tests only top-k mean fusion (Abdollah Pour et al., 2023)	Systematic comparison of max, top-k mean, and rating-weighted fusion, crossed with three sentence encoders

6.4 Unique Features and User Benefits

The system is explainable by design: rather than a bare ranked list, users see the specific review sentences that justify each recommendation. It is built to match underlying query intent rather than exact keywords, which is precisely what the indirect, negation, and contradictory query categories in RIRD are designed to test. The same multilingual pipeline serves both the English evaluation and the Thai demonstration

without a separate translation step, and the encoder-by-fusion comparison grid gives transparency into exactly which modelling choice — the encoder or the aggregation function — drives any improvement in ranking quality, a level of methodological clarity not typically present in Kaggle-style restaurant recommendation notebooks.

CHAPTER 7

PRELIMINARY RESULTS

To confirm the feasibility of the proposed approach before full-scale experimentation, we ran initial exploratory analysis on both the English (RIRD) and Thai (Wongnai, Thailand Foods, M-ABSA) datasets, and additionally implemented and evaluated the two baseline models identified in Chapter 6 on RIRD.

7.1 RIRD: Feasibility as the Evaluation Backbone

The review file contains 29,168 reviews across 50 Toronto restaurants with no missing values in the fields used downstream and no duplicate rows. Star ratings skew positive, consistent with the source dataset's restriction to restaurants with average rating ≥ 3 (Figure 7.1). Review length has a long right tail with a median of approximately 94 words per review (Figure 7.2), confirming that sentence-level splitting is necessary before encoding rather than treating each review as a single unit.

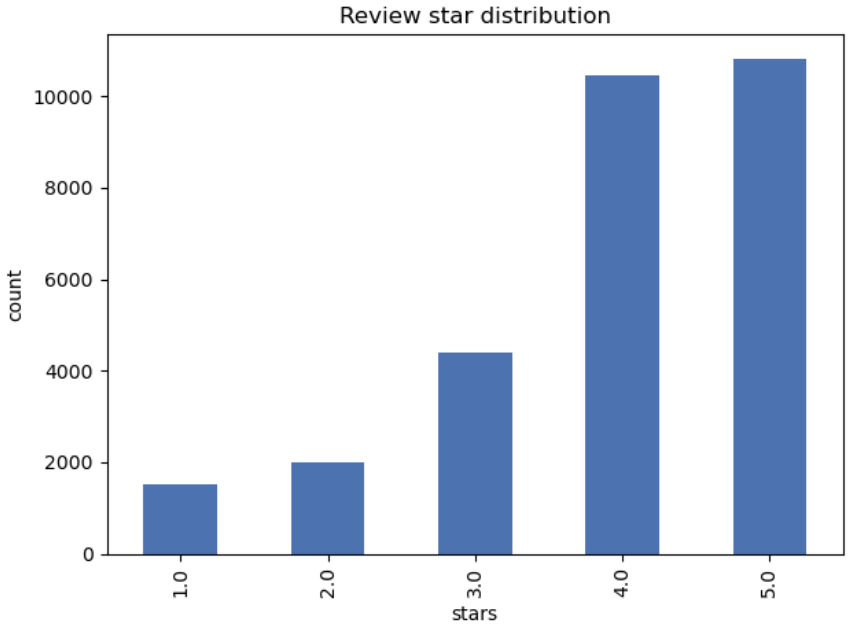


Figure 7.1: Distribution of review star ratings in RIRD.

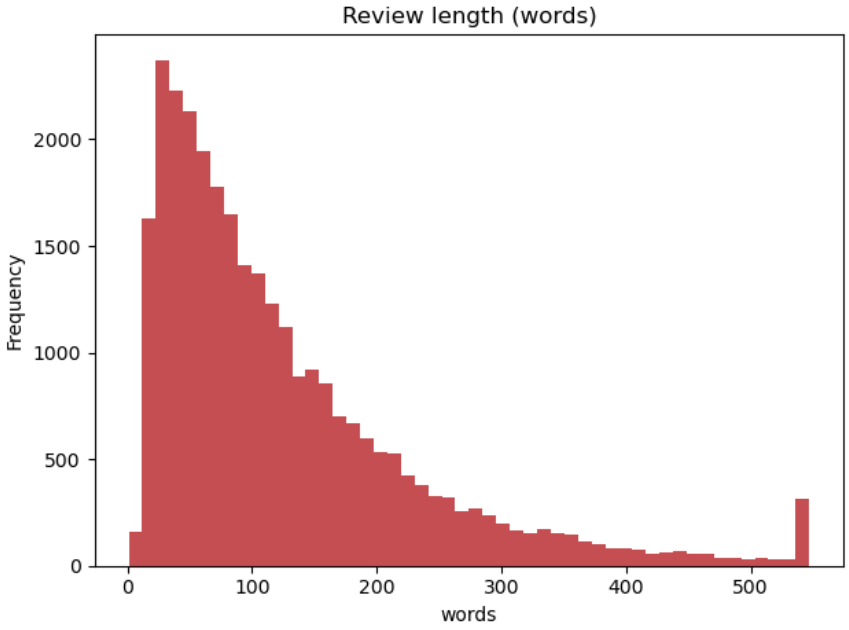


Figure 7.2: Distribution of RIRD review length in words (clipped at the 99th percentile).

A random sample of 5,000 reviews was screened with a Unicode-range script detector; 100% of sampled review text and all 100 PMD queries are Latin-script English, with no Thai content present in either file. This directly confirms the scope decision in Section 2.3: RIRD is used purely as the labelled English evaluation backbone, and no Thai queries are run against it.

7.2 PMD Relevance Labels: Confirming the Evaluation Protocol is Usable

After removing one exact duplicate row (5,000 rows retained from 5,001 raw), the aggregated relevance label is positive for approximately 27% of the 5,000 query–restaurant pairs (100 queries, averaging 13.5 relevant restaurants per query out of 50), confirming the label imbalance anticipated in Chapter 6 and motivating rank-aware metrics over accuracy. The distribution of how many of the five annotators voted "relevant" on the same pair (Figure 7.3) gives an initial picture of annotator agreement ahead of computing formal inter-annotator statistics ($\kappa \approx 0.528$, as reported in the original RIRD release (Abdollah Pour et al., 2023)).

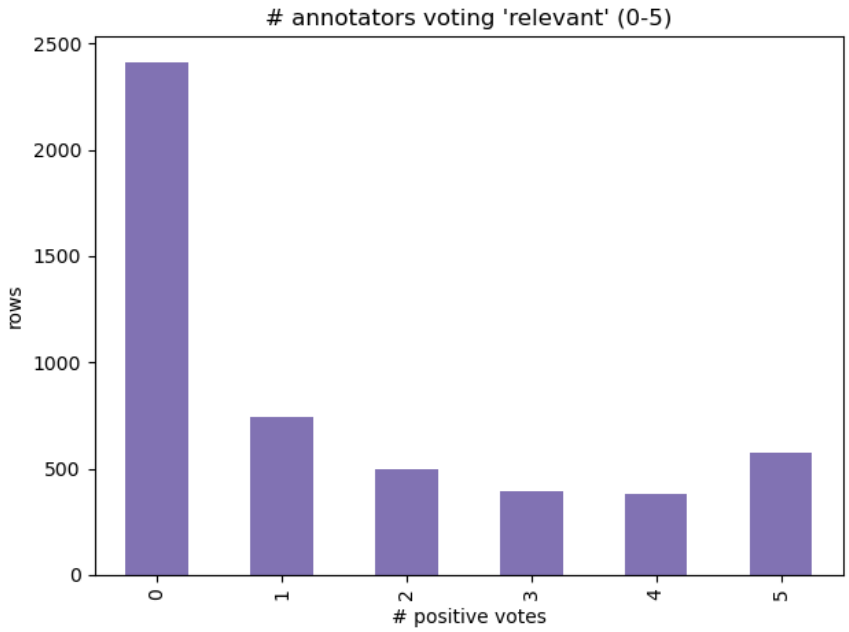


Figure 7.3: Number of annotators (out of five) voting a query–restaurant pair relevant.

7.3 Wongnai: Feasibility and Limits of the Thai Demonstration

The Wongnai corpus consists of 40,000 training and 6,203 test reviews, each with only two fields: review text and a 0–4 star rating (Figure 7.4). Review length is substantially shorter than RIRD's, with a median of approximately 20 words (Figure 7.5), though a long tail remains. A language screen over a 5,000-review sample confirms the corpus is overwhelmingly Thai-script text. Critically, this file contains no restaurant identifier, query field, or relevance label of any kind — direct evidence supporting the scope decision to treat Wongnai as a review/sentence-level demonstration only, rather than a restaurant-ranking evaluation set.

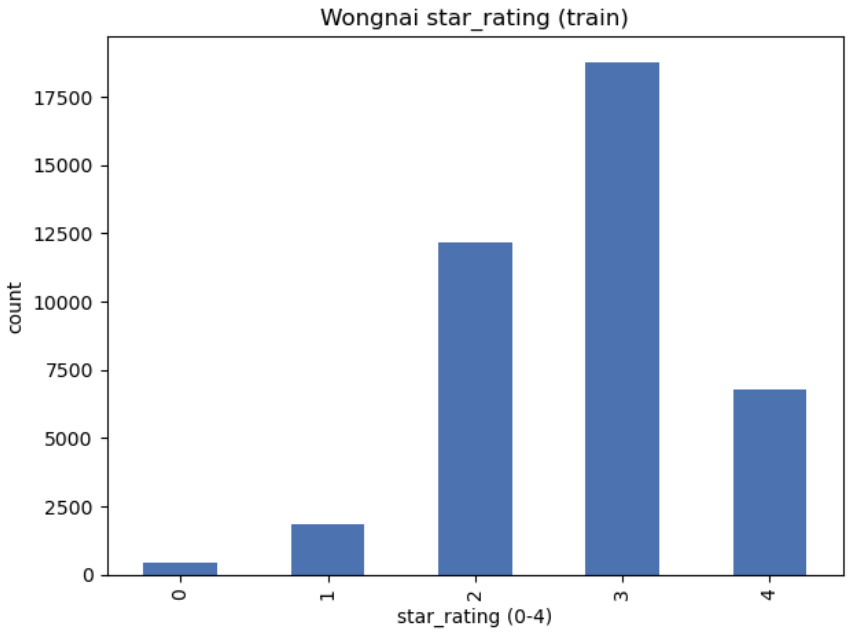


Figure 7.4: Distribution of Wongnai star ratings (0–4 scale).

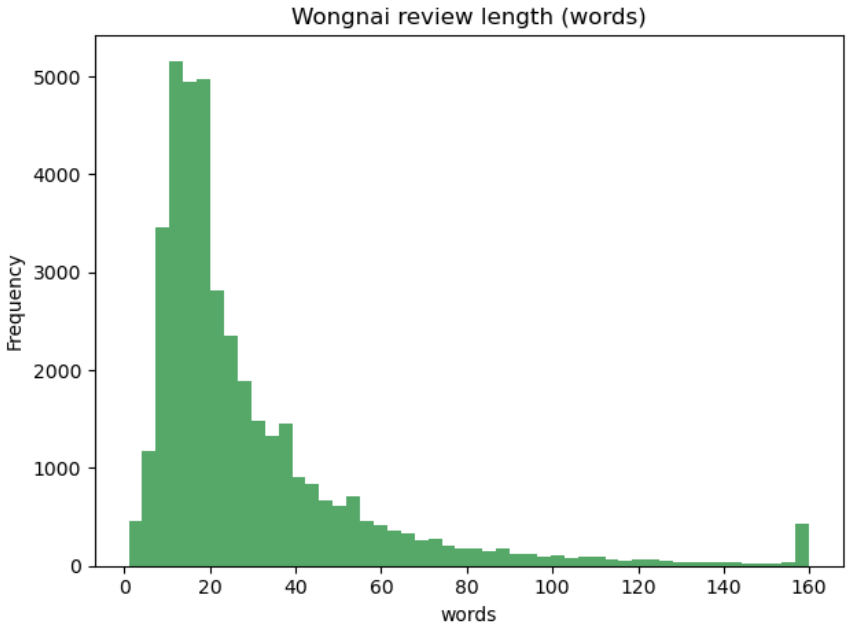


Figure 7.5: Distribution of Wongnai review length in words (clipped at the 99th percentile).

7.4 Thailand Foods and M-ABSA: Feasibility as Supporting Resources

The Thailand Foods gazetteer contains a small number of duplicate English dish names, removed before use; the deduplicated set is dominated by main-course dishes, with region fields heavily populated by *Various*/*Unknown* placeholders rather than a specific Thai region (Figure 7.6). M-ABSA's `restaurant` and `food` domain files were successfully parsed into sentence-level aspect-sentiment tuples; aspect categories are confirmed to be coarse (general food/service/ambience labels) rather than flavour-specific (Figure 7.7), consistent with the design boundary discussed in Section 3.6.

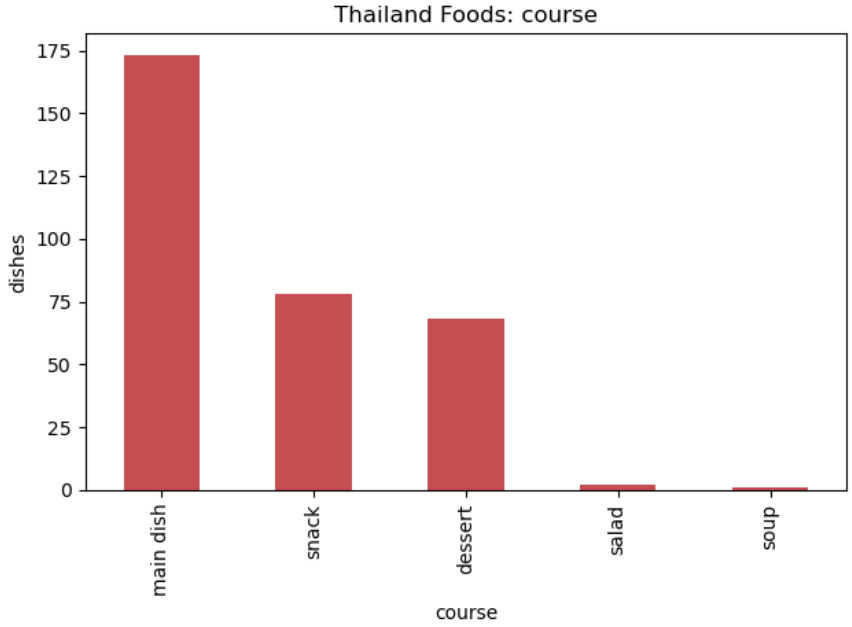


Figure 7.6: Thailand Foods dish counts by meal course.

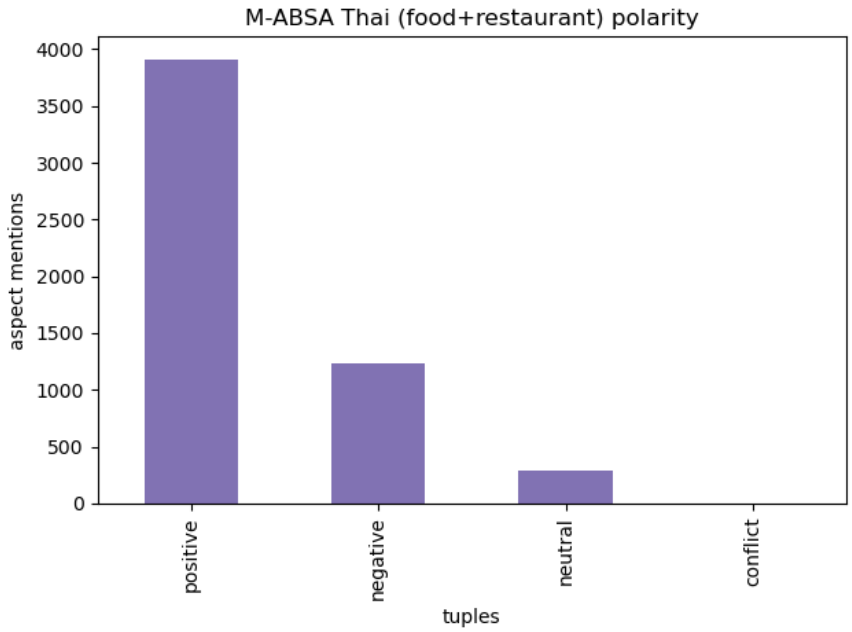


Figure 7.7: Sentiment polarity distribution in M-ABSA Thai (food and restaurant domains).

7.5 Baseline Model Results

Having confirmed the datasets are structurally usable, we implemented and evaluated both baseline models specified in Chapter 6 directly on RIRD, using a shared evaluation harness so that later fusion variants can be inserted into the same comparison without changing the evaluation protocol.

7.5.1 BM25

Reviews were concatenated per restaurant into a single document, tokenised by lowercasing and stripping punctuation, and indexed with `BM25Okapi` over the resulting 50-restaurant corpus. Each of the 100 PMD queries was scored against all 50 restaurants and ranked by BM25 score. Table 7.1 reports the resulting metrics.

Table 7.1: BM25 baseline results on RIRD (100 queries, 50 restaurants).

Metric	Score
MAP	0.5612
MRR	0.7690
nDCG@5	0.6008
Recall@5	0.3156
P@5	0.4620
nDCG@10	0.6211
Recall@10	0.4501
P@10	0.4070

7.5.2 Dense Retrieval Without Fusion

As the second baseline, every review was encoded independently with the multilingual sentence-transformer `paraphrase-multilingual-mpnet-base-v2` (the same encoder planned for the Thai demonstration), and all of a restaurant's review embeddings were mean-pooled into a single 768-dimensional restaurant vector. Queries were encoded with the same model, and restaurants were ranked by cosine similarity between the query embedding and each restaurant's pooled vector. Table 7.2 reports the resulting metrics.

Table 7.2: Dense (no-fusion) baseline results on RIRD (100 queries, 50 restaurants).

Metric	Score
MAP	0.5880
MRR	0.7701
nDCG@5	0.6287
Recall@5	0.3485
P@5	0.5220
nDCG@10	0.6448
Recall@10	0.4758
P@10	0.4450

7.5.3 Cross-Model Comparison

Table 7.3 and Figure 7.8 summarise both baselines side by side. Mean-pooled dense retrieval already outperforms BM25 on every reported metric — most notably MAP (+0.027), P@5 (+0.060), and Recall@10 (+0.026) — despite using no sentence-level fusion at all and no task-specific training. This is consistent with the expectation set in Chapter 6 that dense retrieval should outperform lexical matching on RIRD's free-text preference queries, and it establishes a non-trivial floor for the proposed sentence-level fusion method to improve upon: any fusion variant that fails to beat mean-pooled dense retrieval would indicate the fusion step itself is not adding value.

Table 7.3: Cross-model comparison on RIRD.

Model	MAP	MRR	nDCG@5	Recall@5	P@5	nDCG@10	Recall@10	P@10
BM25	0.5612	0.7690	0.6008	0.3156	0.4620	0.6211	0.4501	0.4070
Dense (no fusion)	0.5880	0.7701	0.6287	0.3485	0.5220	0.6448	0.4758	0.4450

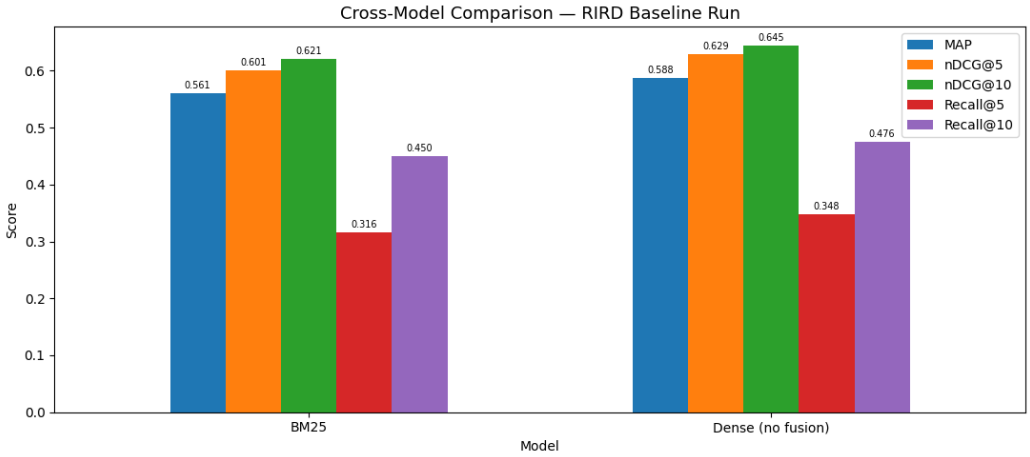


Figure 7.8: Cross-model comparison of BM25 and dense (no-fusion) baselines on RIRD across MAP, nDCG@5/10, and Recall@5/10.

Random ranking and rating-only sorting, the two remaining baselines specified in Chapter 6, together with the full encoder–fusion comparison grid, are left for the full experimental phase of the project; the results above confirm the evaluation harness, qrel construction, and both dataset pipelines are working end-to-end on real data.

7.6 Summary

These initial findings confirm the project's core scope decision and establish feasibility on two fronts. First, RIRD is structurally complete for the intended fusion-comparison experiment (labelled, English, restaurant-grouped, with sufficient review volume per restaurant), while Wongnai, though rich in Thai review text, structurally cannot support anything beyond a sentence-level demonstration without additional restaurant-linking metadata. Second, the full evaluation pipeline — from raw CSV loading through qrel construction, ranking, and metric computation — runs end-to-end on RIRD, and the two implemented baselines already show measurable separation (dense retrieval outperforming BM25 across all eight metrics), giving early evidence that the evaluation protocol is sensitive enough to distinguish between retrieval methods. No feasibility blockers were identified in the datasets or evaluation pipeline selected for either the measured evaluation or the qualitative Thai demonstration.

BIBLIOGRAPHY

Abdollah Pour, M. M., Farinneya, P., Toroghi, A., Korikov, A., Pesaranghader, A., Sajed, T., Bharadwaj, M., Mavrin, B., & Sanner, S. (2023). Self-supervised contrastive BERT fine-tuning for fusion-based reviewed-item retrieval. In *Advances in Information Retrieval (ECIR 2023)*, LNCS vol. 13980 (pp. 3–17). Springer. https://doi.org/10.1007/978-3-031-28244-7_1

Devlin, J., Chang, M.-W., Lee, K., & Toutanova, K. (2019). BERT: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of NAACL-HLT 2019* (pp. 4171–4186). https://doi.org/10.18653/v1/N19-1423

Gao, C., Lei, W., He, X., de Rijke, M., & Chua, T.-S. (2021). Advances and challenges in conversational recommender systems: A survey. *AI Open, 2*, 100–126. https://doi.org/10.1016/j.aiopen.2021.06.002

Highgroundbkk. (n.d.). *wongnai_reviews* [Data set]. Hugging Face. https://huggingface.co/datasets/Highgroundbkk/wongnai_reviews

Karpukhin, V., Oğuz, B., Min, S., Lewis, P., Wu, L., Edunov, S., Chen, D., & Yih, W.-t. (2020). Dense passage retrieval for open-domain question answering. In *Proceedings of EMNLP 2020* (pp. 6769–6781). https://doi.org/10.18653/v1/2020.emnlp-main.550

Koren, Y., Bell, R., & Volinsky, C. (2009). Matrix factorization techniques for recommender systems. *Computer, 42*(8), 30–37. https://doi.org/10.1109/MC.2009.263

Kostric, I., Balog, K., & Radlinski, F. (2024). Generating usage-related questions for preference elicitation in conversational recommender systems. *ACM Transactions on Recommender Systems, 2*(2), Article 12, 1–24. https://doi.org/10.1145/3629981

Lowphansirikul, L., Polpanumas, C., Jantrakulchai, N., & Nutanong, S. (2021). WangchanBERTa: Pretraining transformer-based Thai language models. *arXiv:2101.09635*.

Multilingual-NLP. (n.d.). *M-ABSA* [Data set]. Hugging Face. https://huggingface.co/datasets/Multilingual-NLP/M-ABSA

Pontiki, M., Galanis, D., Pavlopoulos, J., Papageorgiou, H., Androutsopoulos, I., & Manandhar, S. (2014). SemEval-2014 Task 4: Aspect based sentiment analysis. In *Proceedings of the 8th International Workshop on Semantic Evaluation (SemEval 2014)* (pp. 27–35). https://doi.org/10.3115/v1/S14-2004

Reimers, N., & Gurevych, I. (2019). Sentence-BERT: Sentence embeddings using Siamese BERT-networks. In *Proceedings of EMNLP-IJCNLP 2019* (pp. 3982–3992). https://doi.org/10.18653/v1/D19-1410

Robertson, S., & Zaragoza, H. (2009). The probabilistic relevance framework: BM25 and beyond. *Foundations and Trends in Information Retrieval, 3*(4), 333–389. https://doi.org/10.1561/1500000019

Sodchun, P. (n.d.). *Foods in Thailand* [Data set]. Kaggle. https://www.kaggle.com/datasets/ponthakornsodchun/foods-in-thailand

Zhang, Y., & Chen, X. (2020). Explainable recommendation: A survey and new perspectives. *Foundations and Trends in Information Retrieval, 14*(1), 1–101. https://doi.org/10.1561/1500000066